NWPU-Captions Dataset and MLCA-Net for Remote Sensing Image Captioning

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***Abstract*— Recently, the burgeoning demands for captioning-related applications have inspired great endeavors in the remote sensing community. However, current benchmark datasets are deficient in data volume, category variety, and description richness, which hinders the advancement of new remote sensing image captioning approaches, especially those based on deep learning. To overcome this limitation, we present a larger and more challenging benchmark dataset termed NWPU-Captions is available at https://github.com/HaiyanHuang98/NWPU-Captions. NWPU-Captions contains 157 500 sentences, with all 31 500 images annotated manually by seven experienced volunteers. The superiority of NWPU-Captions over current publicly available benchmark datasets not only lies in its much larger scale but also in its wider coverage of complex scenes and the richness and variety of describing vocabularies. Furthermore, a novel encoder–decoder architecture, multilevel and contextual attention network (MLCA-Net), is proposed. MLCA-Net employs a multilevel attention module to adaptively aggregate image features of specific spatial regions and scales and introduces a contextual attention module to explore the latent context hidden in remote sensing images. MLCA-Net improves the flexibility and diversity of the generated captions while keeping their accuracy and conciseness by exploring the properties of scale variations and semantic ambiguity. Finally, the effectiveness, robustness, and generalization of MLCA-Net are proved through extensive experiments on existing datasets and NWPU-Captions.**

***Index Terms*— Contextual attention, NWPU-Captions, remote sensing image captioning.**

I. Introduction

RECENT incredible advancements in deep learning enable a better understanding of remotely sensed data. Although great achievements have been made in conventional object-level visual tasks related to detection, segmentation, and annotation [1], [2], [3], [4], semantic-level tasks such as image captioning remain a tough issue. Captioning for remote sensing images means accurately summarizing the complex scene of a specific image with coherent and compendious natural language descriptions. It not only requires a computer to simulate human vision to recognize dominant objects and explore the relationship between different objects but also requires it to interpret the recognized image content with a sentence that is characterized by clear and accurate syntax and diverse vocabularies. Since remote sensing image captioning provides more specific and semantic-level information than the traditional visual task, it receives wider attention in applications such as cross-modal retrieval, disaster warning, and military intelligence generation [5], [6], [7].

Image captioning produces textual descriptions for a given image in an automatic way, which bridges the field of computer vision and natural language processing. In recent years, continuous endeavors have been devoted to natural image captioning, and various approaches have been proposed [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [39], [40], [41], [42], [43], [44]. Earlier research usually adopts traditional means, i.e., template-based methods [39] or retrieval-based methods [40], to generate descriptive sentences for a certain image. While template-based methods use a two-step object detection and slot-filling process to generate a logically right but rigid sentence, retrieval-based methods search similar images first and then rank textual descriptions for those similar images to obtain a sentence for the query image. With the boom of artificial intelligence technology, researchers turn to applying deep learning to tackle captioning tasks. Comparatively, deep-learning-based methods [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26] usually employ an encoder–decoder architecture, which extracts image features at the encoder end and translates this representation to a descriptive language at the decoder end. Encouraged by the powerful capabilities in feature representation and strong ability to handle text sequence, convolutional neural network (CNN) and recurrent neural network (RNN)/long short-term memory (LSTM) architectures overcome the limitations of traditional methods. They have become the mainstream architectures of image captioning.

Currently, considerable endeavors have been devoted to promoting benchmark datasets and new approaches for captioning

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in the remote sensing field. For example, Qu *et al.* [27] pioneered a deep multimodal neural network model and two benchmark datasets, Sydney-Captions [27] and UCM-Captions [27]. Their model combined different CNNs with RNN/LSTMs to improve the performance. Afterward, Lu *et al.* [28] analyzed and summarized the challenges of remote sensing image captioning and constructed a larger benchmark dataset, RSICD [28], to generate more accurate and flexible descriptions. Subsequently, a variety of improved approaches were proposed under the consideration of the intrinsic characteristics of remote sensing images. Among them, typical and effective solutions are to apply various kinds of attention mechanisms to the vanilla encoder–decoder architecture [29], [30], [31], [32], [33], [34], [35], and some of the methods [34], [35] achieved promising performance.

However, further development of novel captioning approaches in the remote sensing field is restrained by the deficiency of existing datasets. The limitations could be summarized in three aspects.

1) Large volumes of labeled training data are necessary for deep-learning-based methods, but current datasets are relatively small. Both the number of images and the categories are insufficient. Specifically, UCM-Captions [27], Sydney-Captions [27], and RSICD [28] include 10 500 sentences of 2100 sentences of 613 images and 54 605 sentences of 10 921 images.
2) Some complex and fine scenes, such as wetland and rectangular farmland, are not covered, which results in poor performance in understanding and describing complex image scenes.
3) The diversity of descriptions is insufficient, and for instance, 30 272 of the 54 605 sentences of RSICD are generated through randomly duplicating. Therefore, a new large-scale benchmark dataset, which contains more images and classes in diverse and richer descriptions, is of great significance.

Motivated by this demand, we present the NWPU-Captions dataset. The proposed NWPU-Captions is constructed based on NWPU-RESISC45 [36], consisting of 31 500 images with expansive coverage of 45 classes. The reason why we choose NWPU-RESISC45 is its merit in image number and scene classes, showing large variations and high within-class diversity/between-class similarity. To construct NWPU-Captions, each image of NWPU-RESISC45 is annotated manually in five different sentences according to certain rules, and the total number of descriptions reaches 157 500.

As a matter of fact, the task of remote sensing image captioning goes far beyond datasets. There remain huge challenges in generating precise semantic descriptions from remote sensing imageries with complex backgrounds, diverse land categories, weak textures, and varied viewpoints or scales. In a brief investigation, the following factors are particularly prominent.

1) *Scale Discrepancy Between Data Sources:* It refers to the scale-related ambiguity with which the same kind of ground objects at different spatial resolutions may present different semantics, e.g., "airplane" under high spatial resolution and "airport" under low spatial resolution.
2) *Category Ambiguity Over Wide-Range Regions:* Typical remote sensing scenes may contain multiple land categories, so it is hard to describe the compound regions (e.g., urban–rural junction areas) where there are heterogeneous categories.

To address these challenges, we further put forward a novel encoder–decoder architecture, namely, multilevel and contextual attention network (MLCA-Net), for remote sensing image captioning. In MLCA-Net, while different levels of features, including features of specific spatial locations and specific scales, are adaptively fused through a multilevel attention module, the latent contextual information of the image is explicitly explored through a contextual attention module. The former can enforce the salient visual representations through spatial and channelwise attention (SCA) [61], whereas the latter pays more attention to semantic associations through the attribute attention mechanism [29] to overcome the difficulties of semantic description suffering from scale variations. To mitigate ambiguity issues, a visual sentinel [60] is additionally introduced to the LSTM-based decoder to adaptively align extracted visual features and generate descriptive words from the semantic level, which is achieved by determining the attention weight required for visual features to predict the next word.

The major contributions of this article are from three aspects.

1) A by far largest remote sensing image captioning benchmark dataset, NWPU-Captions, is presented. Compared with current benchmark datasets, the advantages of NWPU-Captions not only lie in its much larger scale but also, more importantly, in its wider coverage of complex scenes and the richness and variety in vocabulary and sentences. NWPU-Captions serve as the baseline for developing captioning and cross-modal retrieval approaches.
2) A novel encoder–decoder architecture, named MLCA-Net, is proposed to mitigate scale discrepancy and category ambiguity derived from multisource remotely sensed data. Mainly, multilevel attention and contextual attention are exploited for aggregating different levels of features adaptively to tackle scale variations. Besides, the LSTM-based decoder adaptively aligns visual features and semantic descriptors, thereby alleviating the problem of ambiguity. Compared with the existing attention-based encoder–decoder architectures, MLCA-Net focuses more on the context mining of complex scenes and the semantic alignment between visual features and attribute information.
3) Comprehensive experiments are carried out on existing benchmark datasets and proposed NWPU-Captions to evaluate the effectiveness, robustness, and generalization of MLCA-Net.

II. Related Works

Research on extracting semantic information from images and describing the extracted information in natural language

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can be dated back to the beginning of the 21st century. Benefited from the high-speed growth of artificial intelligence and deep learning technology, remarkable achievements have been gained in image captioning. In the following, we separately investigate recent progress on captioning for natural images and remote sensing images.

A. Datasets

1) UCM-Captions: UCM-Captions [27] is the earliest publicly available dataset for remote sensing image captioning tasks. It was constructed based on the baseline dataset of remote sensing image retrieval and classification, UCMD [37]. UCMD contains 2100 images from 21 categories, and the image size is 256 × 256 pixels. UCM-Captions is built by describing each image of UCMD with five different sentences and hence contains 10 500 descriptions in total. In the five sentences, while different phrasings are used, the meaning keeps the same.

2) Sydney-Captions: Sydney-Captions [27] was constructed based on the Sydney dataset [38]. All of the images were cropped from 18 000 × 14 000 pixels large images of Sydney produced on Google Earth. The cropped 500 × 500 pixels sized images belonged to seven scene categories, and each image was annotated by five different descriptions. Thus, Sydney-Captions contains 3065 sentences in total. Similar to UCM-Captions, each image of Sydney-Captions is described by five sentences that are composed of different phrases and words but have the same meaning.

3) RSICD Dataset: RSICD dataset [28] is larger than the aforementioned two datasets. All images collected from different sources were cropped to the pixel size of 224 × 224 with various spatial resolutions. RSICD contains 10 921 images from 30 categories, and the total number of descriptions was 24 333 originally. Finally, it was extended to 54 605 sentences by randomly duplicating the existing sentences for those images that have not five different sentences.

As the earlier publicly available remote sensing dataset, UCMD got extensive applications in image retrieval and scene classification. However, its categories and the images per category are relatively insufficient. The Sydney dataset is produced based on Sydney, NSW, Australia, from Google Earth, and only seven classes of scene are included, which is really limited. Comparatively, RSICD contains more categories and more images per category than UCM-Captions and Sydney-Captions. The deficiency of RSICD is that many images only have one or two descriptions. For example, 4853 of 10 921 images were annotated by only one sentence. Among the total 54 605 sentences of RSICD, 24 333 sentences are annotated manually, while other 30 272 sentences are generated through a simple process of random duplicating. These kinds of captioning result in a lack of diversity and richness in vocabulary and sentences as well, which is insufficient for the development and evaluation of new deep-learning-based approaches that demand a massive amount of labeled data. Hence, a larger and more challenging dataset that provides more images and categories annotated with richer descriptions as possible is urgently needed.

B. Natural Image Captioning

1) Traditional Methods: Earlier, template- and retrieval-based methods were generally exploited in image captioning.

Template-based methods generate descriptive sentences for a given image through a two-step process of object detecting and template filling. Specifically, these methods utilize object detection to match the extracted image features (i.e., object, attribute, relationship, and action) into the preset template, hence generating a simple description for an image. Li *et al.* [39] detected objects and their visual attributes and spatial relationship first and then fused the selected phrases to compose image descriptions. Venugopalan *et al.* [41] presented the novel object captioner (NOC), which leverages labeled images and semantic knowledge to include "outside" objects that were absent in current image-caption datasets into the generated captions. Lu *et al.* [18] presented a two-stage end-to-end differentiable architecture, Neural Baby Talk, by reconciling conventional slot filling with object detectors. The merit of template-based methods is that they are intuitive and easy for manipulation and understanding. However, although the correctness of grammar and syntax are guaranteed, the sentences generated are inflexible and rigid, which leads to limited-expression ability.

Retrieval-based methods utilize sentences derived from similar images to describe the query image. These kinds of methods need retrieval first, and the generation of corresponding sentences is based on the similarity calculation between the query image and candidate sentences. Gupta *et al.* [42] extracted linguistically motivated phrases from human-generated descriptions for an image dataset. Then, the phrases present in its neighboring images were ranked based on image similarity and integrated to construct triples for the final sentence generation. Farhadi *et al.* [43] proposed a symmetric approach to link images to sentences. They built an intermediate space, namely, meaning space, to get different projections from image space and sentence space. The final caption is obtained by scoring the similarity in meaning space. Hodosh *et al.* [44] treated captioning task as a ranking procedure and managed to map visual and textural information into a public space for similarity calculation. Retrieval-based methods are advantageous in which the generated sentences are relatively flexible and diverse, while the disadvantage is that the performance of sentence generation heavily relies on retrieval results.

2) Deep-Learning-Based Methods: Deep-learning-based methods, which usually adopt CNN and RNN/LSTM to extract features and generate descriptions, are boosted by the fast growth of deep learning technology. Pioneering works with regard to the encoder–decoder architecture include the multimodal recurrent neural network (m-RNN) model [8] and the neural image captioning (NIC) model [9]. In subsequent work, various improvements have been made to the fundamental encoder–decoder architecture for different task requirements. For example, attention mechanisms [10], [11], [12] are exploited to meet the demand of local attention visual tasks, a shared multimodal space module [13] is used for multimodal conversion visual tasks, and generative adversarial network

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(GAN) [14] and reinforcement learning (RL) [15], [16] are introduced to deal with insufficient training samples.

The variants of the encoder–decoder architecture can also be categorized into encoder-based variants and decoder-based variants. Encoder-based variants use a multi-instance training word detector to extract keywords [17] and introduce object detection as the encoder input [18], [19]. Decoder-based variants include neural networks for extracting the sentence templates [18], stylization of the decoding process [20], hierarchical and layered decoding [21] [22], external information through the knowledge graph [23], multimodel joint training [24], extended LSTM model (g-LSTM), semantic information input [25], and multilayer LSTMs model [26].

Due to its superiority in the generation of sentences in semantical accuracy, flexibility, and variety, deep-learning-based methods have received extensive attention of the community. They have become mainstream architectures in the image captioning field.

C. Remote Sensing Image Captioning

The booming demands for remote sensing image captioning-related applications stimulate enormous enthusiasm and efforts on this semantic-level visual task. Related work could be traced back to 2016 [27]

First, Qu *et al.* [27] proposed a multimodal encoder–decoder architecture to interpret remote sensing images with high resolution from a semantic level. In their model, features extracted from CNN are combined with textural descriptions by RNN/LSTMs to produce final sentences. They proved the effectiveness of their architecture on their proposed datasets, UCM-Captions and Sydney-Captions.

Subsequently, the third dataset, RSICD [28], was created to overcome the limitation of the former two datasets in data volume and categories. Since then, continuous research has been conducted on different aspects such as metric learning, scale differences, and sentence ambiguity. Specifically, a collaborative semantic metric learning framework (CSMLF) [45], which measures the semantic distance between images and descriptive statements via semantic embedding, was proposed. A multiscale cropping training mechanism [46] was presented to enhance the generalization performance of the model while extracting more fine-grained information. A denoising-based multiscale feature fusion (DMSFF) [47] was proposed for the aggregation of multiscale features. A truncation cross-entropy (TCE) loss [48] was suggested to mitigate the phenomenon of overfitting caused by a cross-entropy loss. A retrieval topic recurrent memory network (RTRMN) [49] was proposed to solve the ambiguity that resulted from five statements describing the same input image. Besides, an explainable word–sentence framework [50] was presented to overcome the deficiency of the black-box model. The authors believed that the image-word-to-sentence model of the word–sentence model is more in accordance with the intuition and logicality of human understanding of an image than the encoder–decoder framework. A novel image captioning framework [51] combined beam-search caption generation with image retrieval was proposed. A novel summarization-driven RS image captioning (SD-RSIC) approach [62], which combined standard caption generation with ground-truth (GT) caption summarization, was proposed. A novel CNN-SVM network [63] that efficiently shortens training and inference times was proposed. Those efforts boost the development of remote sensing image captioning greatly.

Considerable concerns have been paid to the adoption of attention mechanisms into captioning approaches. Various kinds of attentional modules were exploited on either encoder end or decoder end to meet the requirements of local attention visual tasks. An attribute attention-based architecture [29] was proposed in which different regions of input images were allotted different weights. A multilabel attribute graph CNN [30] was presented to address the scale differences and spatial characteristics of remotely sensed images. A multilevel attention model [31], which combined three different kinds of attention to imitate the human attention mechanism, was presented. The authors also did a lot of work to modify the word errors, grammatical errors, and inappropriate descriptions for UCM-Captions, Sydney-Captions, and RSICD. A visual aligning attention (VAA) model [32] optimized the attention layer of the decoder with a well-designed attention loss function. In VAA, nonvisual words were filtered out through a visual vocabulary for the purpose of eliminating their influence on the attention layer. A label-attention mechanism (LAM) [33] utilized label information to guide the calculation of attention masks. Multiscale attention (MSA) and multifeat attention (MFA) were put forward to tackle the multiscale problem of remotely sensed images. A persistent memory mechanism (PMM) [53] was proposed to solve the problem of information loss derived from limited storage capacity under the condition of the long caption. A new structured attention-based method [34] was presented to overcome the deficiency of standard attention-based captioning methods, which usually neglect structured information characteristics of remote sensing images. In this method, the learned discriminative descriptions and produced pixelwise segmentation were fed to the network and to be trained with the captioning information. A novel framework [35] that integrates static and multiscale features to generate robust features was proposed. In this framework, recurrent attention and semantic gate were exploited to enhance the performance. Two methods, MSA and MFA [52], which applied multihead attention mechanism and target-level feature extraction to tackle multiscale problems, were proposed. For a systematic survey of remote sensing image captioning, readers can refer to [64].

III. NWPU-Captions: A Large-Scale Dataset for Remote Sensing Image Captioning

The deficiency of existing benchmark datasets in the number of images and scene types and diversity of descriptions hinders the advancement of novel captioning approaches. Therefore, we present the NWPU-Captions dataset.

NWPU-Captions is constructed based on the NWPU-RESISC45 dataset [36], which is one of the largest benchmark datasets currently available and has served as a baseline for classification tasks in remote sensing. The images of NWPU-RESISC45 are collected from Google Earth, and the total

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number of images is 31 500, which covers 45 scene classes. Each class contains 700 images, and the image size is 256 × 256 pixels. The challenge of NWPU-RESISC45 not only lies in its number of images and scene classes but also lies in its large variations and high within-class diversity/between-class similarity characteristics derived from data acquisition discrepancy. Specifically, the spatial resolutions of NWPU-RESISC45 range from about 30 to 0.2 m/pixel for most scene classes except for several classes with lower spatial resolutions such as island, lake, mountain, and snowberg. What is more, much finer categories with a high degree of semantic overlapping are related, such as circles and rectangular farmland, commercial and industrial areas, and basketball and tennis courts.

The construction of NWPU-Captions is done by seven volunteers who major in remote sensing. To guarantee the diversity of vocabulary and syntax, each image of NWPU-RESISC45 is annotated manually by five different sentences, and hence, NWPU-Captions dataset contains 157 500 originally different sentences. Taking the annotated rules provided in [28] as a reference, we follow the instructions below in the construction of NWPU-Captions.

1) A sentence should contain at least six words.
2) All prominent objects of the input remote sensing image should be described.
3) Ambiguous words and syntax should not be used.
4) Vague words, such as large, small, short, and tall, should not be used in a sentence.
5) Directional words, such as up, down, left, and right, should not be used.
6) Fresh vocabularies and flexible syntax are encouraged to use.

Table I comprehensively compares existing benchmark datasets and NWPU-Captions in terms of the number of images, category, vocabulary, the total number of description sentences, and the balance of categories. Some example sentences are shown in Fig. 1. It shows that the NWPU-Captions dataset provides more detailed information for complex scenes, in which vocabularies are more plentiful and syntactical structures are more diverse.

IV. MLCA-Net: Multilevel and Contextual Attention Network

We propose a new encoder–decoder architecture, MLCA-Net, focusing on the scale discrepancy and category ambiguity issues in remote sensing images. MLCA-Net is composed of two main parts, as shown in Fig. 2. At the encoder end, the multilevel attention module and the contextual attention module are embedded to extract multilevel and contextual representation, respectively. The multilevel attention module adopts the SCA proposed in [61]. While spatial attention deals with the aggregation of pixels in the adjacent regions that have higher semantic correlation, channel attention copes with the screening of background noises and redundant information to further emphasize the salient objects in remote sensing images. The contextual attention module, employing attribute attention mechanism [29], explicitly attends to correlative feature patches at distant spatial locations, which facilitates understanding an image as a whole through analyzing the spatial coherency of different parts. At the decoder end, extracted multilevel image features and contextual features are fed into LSTM as input. Then, a visual sentinel [60] is introduced to adaptively determine whether it needs to attend to the image to predict the next word. The adaptive alignment of extracted visual features with generated descriptive words is helpful in improving the discriminative capability of predicted description sentences.

A. Attention Module in Encoder

The attention mechanism has been successfully adopted in various visual tasks, including image captioning. In MLCA-Net, different features (namely, features of specific spatial locations and features of specific scales) are adaptively fused through multilevel attention, and potential contextual information of the input image is explicitly explored through a contextual attention module, as shown in Fig. 3.

1) Multilevel Attention: Remote sensing observations are captured from various sensors with different spatial resolutions equipped on airborne or spaceborne platforms, which generates scale discrepancy for the same kind of ground objects and may lead to scale-related ambiguity for understanding the semantics of the images. For CNNs, the feature maps from deeper layers contain higher semantic level features, whereas feature maps from shallower layers render more detailed features.

The main idea of the multilevel attention module could be described as follows. First, features at different levels are concatenated to generate multiscale deep representations, which are then embedded in different channels for fusing features. Afterward, channel and spatialwise attention [61] is adopted to select features of specific scales and spatial locations adaptively. We take VGG16 as the backbone network by comprehensively considering accuracy and training time.

Assuming that I represents the input image, the intermediate features extracted from the fully connected layer, $\text{conv}4_3$ and $\text{conv}5_3$, are depicted as F_{fc}, F_{L1}, and F_{L2}, respectively.

$$F_{\text{fc}} = \text{CNN}_{\text{fc}}(I) \tag{1}$$

$$F_{L1} = \text{CNN}_{\text{conv4}}(I) \tag{2}$$

$$F_{L2} = \text{CNN}_{\text{conv5}}(I). \tag{3}$$

The output feature maps F_L of different convolutional layers are used as the basic features of the image. After upsampling and average pooling, the feature map size of all images is uniformly set to $7 \times 7 \times 2048$, which means that the feature vector dimension corresponding to each image area is set to 2048. Then, the multilevel feature F_{ml} of the image is obtained through merging the output of feature maps from different layers with spatial attention weight α and channel attention weight β

$$V_{\text{ml}} = \text{concat}(F_{L1}, \text{upsample}(F_{L2})) \tag{4}$$

$$\alpha = \text{softmax}(W_i \tanh((W_s V_{\text{ml}} + b_s) \oplus W_{\text{hs}} h_{t-1}) + b_i) \tag{5}$$

$$\beta = \text{softmax}\left(W_i' \tanh((W_c \oplus V_{\text{ml}} + b_c) \oplus W_{\text{hc}} h_{t-1}) + b_i'\right) \tag{6}$$

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TABLE I
COMPARISON OF EXISTING THREE CAPTIONING DATASETS WITH OUR NWPU-CAPTIONS

Dataset	UCM-Captions[27]	Sydney-Captions[27]	RSICD[28]	NWPU-Captions
Number of images	2,100	613	10,921	31,500
Image size	256×256	500×500	224×224	256×256
Spatial resolution	around 0.3 m	0.5 m	different resolutions	~30-0.2m
Captioning mode	5 sentences per image	5 sentences per image	1-5 sentences per image	5 sentences per image
Number of sentences	10,500	3,065	24,333 originally, 54,605 after duplicating extending	157,500
Number of distinct vocabularies	285	169	2,632	3,149
Balanced	Balanced	Unbalanced	Balanced	Balanced
Annotation	Same person	Same person	Different Person	Different Person
Scene classes	21classes: agricultural, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium density residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks and tennis courts	7 classes: residential, airport, meadow, rivers, ocean, industrial, and runway	30 classes: airport, bare land, baseball field, bridge, center, church, commercial, dense residential, desert, farmland, forest, industrial, meadow, medium residential, mountain, park, school, square, parking, playground, pond, viaduct, port, railway station, resort, river, sparse residential, storage tanks, stadium	45 classes: airplane, airport, baseball diamond, basketball court, beach, bridge, chaparral, church, circular farmland, cloud, commercial area, dense residential, desert, forest, freeway, golf course, ground track field, harbor, industrial area, intersection, island, lake, meadow, medium residential, mobile home park, mountain, overpass, palace, parking lot, railway, railway station, rectangular farmland, river, roundabout, runway, sea ice, ship, snow berg, sparse residential, stadium, storage tank, tennis court, terrace, thermal power station, and wetland.

$$F_{\mathrm{ml}} = \beta * \left(\sum_{i=1}^{P} \alpha_t^i V_{\mathrm{ml}_i} \right) + \sum_{i=1}^{P} \alpha_t^i V_{\mathrm{ml}_i} \tag{7}$$

where F_{ml} is the final multilevel features, α and β, respectively, refer to spatial attention weight and channel attention weight, and V denotes the output of feature maps of the convolutional layer. $W_s \in \mathbb{R}^{k \times C}$, $W_{\mathrm{hs}} \in \mathbb{R}^{k \times d}$, and $W_i \in \mathbb{R}^k$ is a matrix that converts visual feature maps and hidden states into the same dimension.

2) Contextual Attention: Contextual information helps understanding images from semantic levels, which is essential to boost the discriminative ability of image representation. Considering that there exists a certain loss of information along with the propagation process of convolutional layers, MLCA-Net leverages a contextual attention module based on the attribute attention mechanism [29] to fuse visual features from different levels of CNN to learn descriptive sentences.

Specifically, different weights are assigned to different regions of an image through the contextual attention module. The weight γ_{ij} can be figured through the relevancy between the j_{th} feature vector and the $(i-1)_{\mathrm{th}}$ word of the descriptive language, shown as follows:

$$e_{ij} = a\left(s_{i-1}, f_j\right) \tag{8}$$

$$\gamma_{ij} = \frac{\exp\left(e_{ij}\right)}{\sum_k \exp(e_{ik})} \tag{9}$$

$$c_i = \sum_j \gamma_{ij} f_j \tag{10}$$

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UCM-Captions[27]		*This is a dense residential area.* *It is a dense residential area with lots of houses arranged neatly, and a road goes through them.* *There are lots of houses with grey roofs arranged neatly, and a road goes through them.* *Lots of houses are arranged neatly, and some cars are parked on the roadside.* *Lots of houses with grey roofs arranged neatly and some cars parked on the roadside.*
		*Two tennis courts next to a stadium with a road surrounded.* *Two tennis courts are next to a stadium with a road beside.* *There are two tennis courts next to a stadium and surrounded by some plants.* *There are two tennis courts next to a stadium with a road and some plants beside.* *There are two tennis courts next to a stadium with a road beside.*
Sydney-Captions[27]		*Some roads go through the residential area.* *A residential area with houses arranged neatly and a highway beside this area.* *A town with many houses arranged neatly while some roads go through this area.* *A residential area and a wasteland are separated by a highway.* *A residential area and a wasteland are separated by a highway.*
		*Many airplanes with red fuselages are parked neatly at the airport.* *There are some airplanes with red fuselages parked neatly at the airport.* *Some airplanes with red fuselages parked neatly at the airport.* *Some airplanes with red fuselages parked neatly at the airport.* *Many airplanes with red fuselages parked neatly at the airport.*
RSICD[28]		*A huge playground is accompanied by several small fields.* *A large rectangular basketball court and near by a grey road.* *Some buildings and some green trees are around a playground.* *A playground is surrounded by some buildings and green trees.* *A large playground is surrounded by many trees and houses.*
		*Many tall buildings have been built in the central area.* *The central building is surrounded by a large number of densely populated areas.* *The central building is surrounded by a large number of densely populated areas.* *Many tall buildings have been built in the central area.* *Many tall buildings have been built in the central area.*
NWPU-Captions		*The industrial area has some blue workshops and green areas, and some roads go through the industrial area.* *Houses in industrial areas vary in size.* *Neatly planned plants and roads in an industrial area with good greening.* *There are many roads in the industrial area.* *There are some blue buildings and many black buildings in the industrial area.*
		*The roundabout with three exits and entrances is in the residential area, and some clearings are next to the roundabout.* *There are several buildings with swimming pools and some trees, and a large lawn around the roundabout with three roads.* *The roundabout is next to buildings and trees.* *There are many buildings of different shapes and sizes around the roundabout.* *There are many buildings around the roundabout.*

Fig. 1. Examples of the four RS image captioning dataset.

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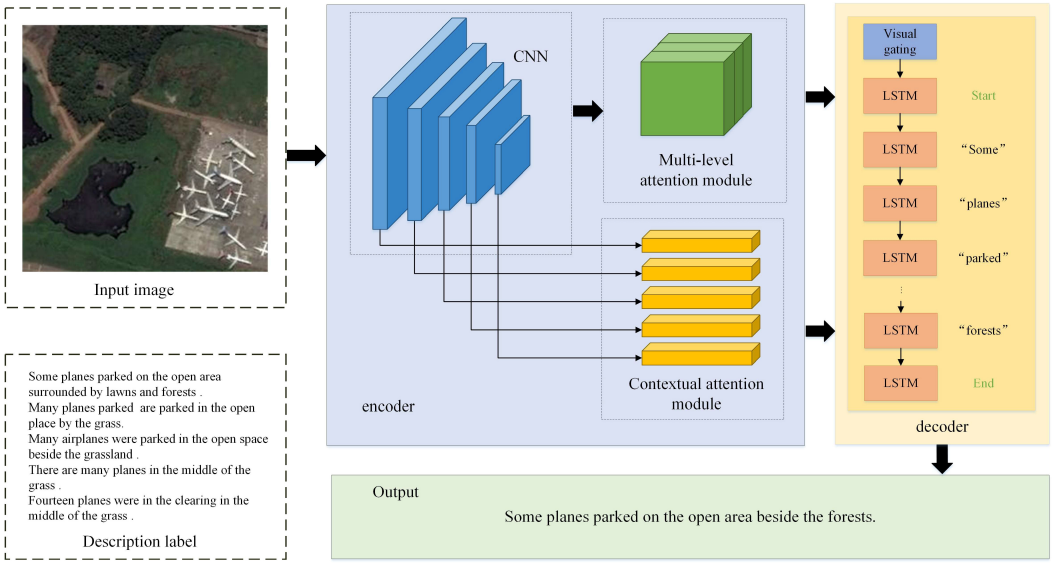


Fig. 2. Overall architecture of MLCA-Net.

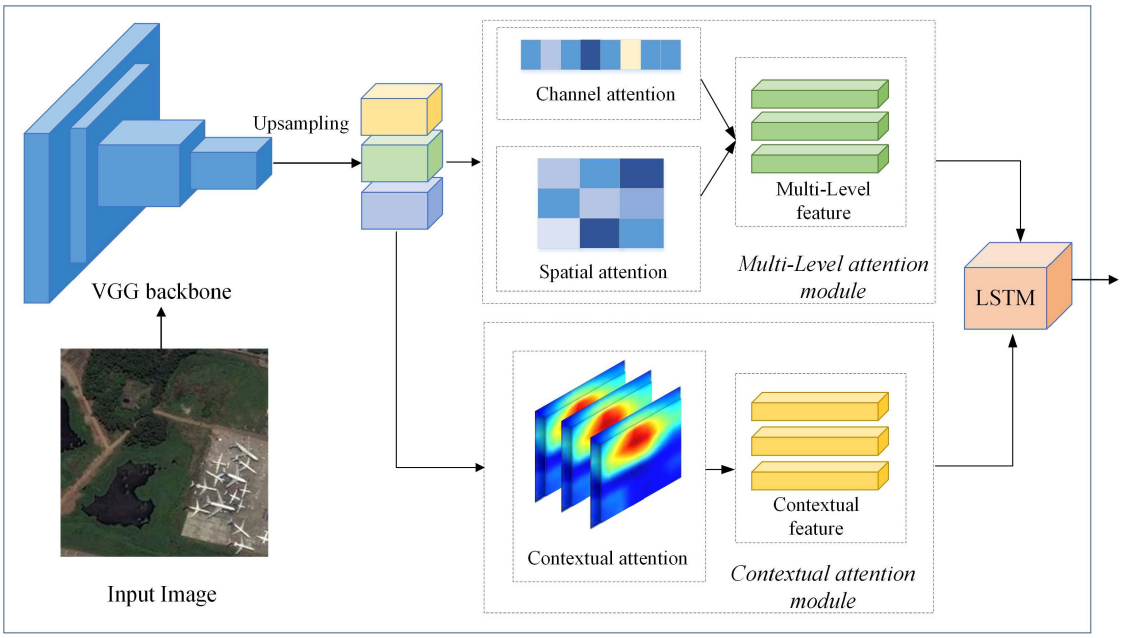


Fig. 3. Outline of the encoder from images to visual features.

where $a(\cdot)$ represents the alignment model, s_{i-1} indicates the $(i-1)$th word of the sentence generated by LSTM, f_j represents the feature vector of the j_{th} region of the image, e_{ij} denotes the similarity of s_{i-1} and f_j, and c_i is an intermediate variable.

Since features of different layer CNNs encode different levels of information, features extracted from the deeper layer pay closer attention to abstract and semantic content, while features extracted from shallow layers embody more specific and detailed information. Also, as the deepening of the convolutional layers, the receptive field increases so that the corresponding feature maps tend to contain global contextual information. However, previous work [10] [28] is more likely to ignore high-level image features and might result in category ambiguity.

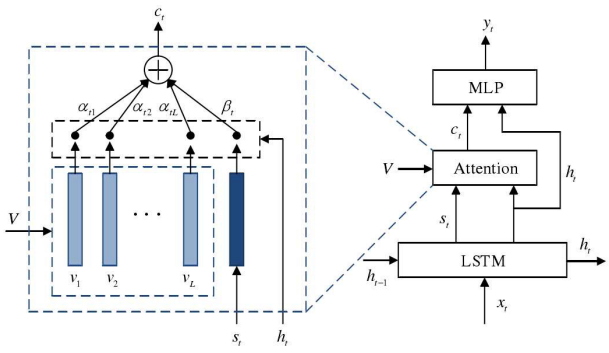


Fig. 4. Visual sentinel model [60] in LSTM-based decoder.

As shown in (8) and (9), the weight γ_{ij} is affected by the output s_i of the image features f_j and SoftMax. The change

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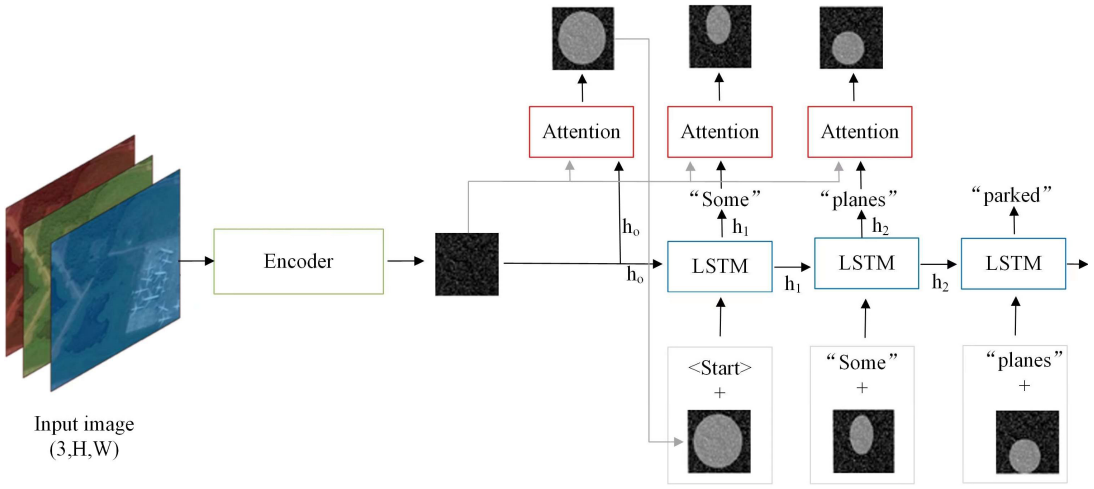


Fig. 5. Outline of the decoder from visual features to description sentences.

of weight τ_{ij} could be implemented by changing the value of e. In MLCA-Net, a constructing method of e is used to generate contextual features F_{con}, which fuses local and global contextual information from different layers of CNN

$$e_{ij} = a(s_{i-1}, W_{\text{sh}}\text{concat}(F_{\text{fc}}, W_c V_{\text{ml}})) \tag{11}$$

$$\tau_{ij} = \frac{\exp(e_{ij})}{\sum_k \exp(e_{ik})} \tag{12}$$

$$F_{\text{con}} = \sum_j \tau_{ij} V_{\text{ml}} \tag{13}$$

where s_{i-1} is the $(i-1)$th word in the description sentence generated by LSTM at the decoder end, W_{sh} is the dimension adjustment factor, W_c is the learnable weight, F_{fc} represents fully connected layer features (fc-CNN), and V_{ml} represents the multilayer visual features obtained by the fusion of multiple CNN convolutional layers.

B. Visual Adaption in LSTM-Based Decoder

The extracted multilevel image features F_{ml} and contextual features F_{con} are then fed into the decoder of MLCA-Net. Next, the label sentence $S = \{S_1, S_1, \ldots, S_L\}$ is used to generate the rank in turn and get the final coherent sentence.

As shown in Fig. 4, a visual sentinel [60] is an adaptive model introduced to the LSTM-based decoder of MLCA-Net, which aims to automatically determine whether it needs to attend to the image to predict the next word through attention weights. Therefore, nonvisual words, such as "a/the" and "nearby," can be excluded to align with extracted image features.

First, the initial hidden state and memory state are created via the generated visual features. Then, a weight for each pixel in the attention network is generated through the visual features and the previous hidden state. The weight reflects the association degree of each region and the word generated previously. Afterward, the weighted average of the previously generated words and visual features is fed back to the decoder for the generation of the next word. As shown in Fig. 5, the process from the visual features to description sentences is a process of visual adaption, which makes semantic alignment between extracted visual features and generated descriptive words.

In MLCA-Net, the input x_t of LSTM is the multilevel features F_{ml} and the contextual features F_{con} obtained at the encoder end. The LSTM-based decoder uses them to readjust the weight of the features and focus on specific areas of the image dynamically, aiming to generate a description sentence reflecting the scale and contextual information. The LSTM is decoded by adding multilevel features F_{ml} and context features F_{con} to all gates. The decoding process could be described with the following formulas:

$$i_t = \sigma\left(W_{ix}x_t + W_{ih}h_{t-1} + W_{ia}F_{\text{ml}_t} + W_{ic}F_{\text{con}t} + b_i\right) \tag{14}$$

$$f_t = \sigma\left(W_{fx}x_t + W_{fh}h_{t-1} + W_{fa}F_{\text{ml}_t} + W_{fc}F_{\text{con}t} + b_f\right) \tag{15}$$

$$o_t = \sigma\left(W_{ox}x_t + W_{oh}h_{t-1} + W_{oa}F_{\text{ml}_t} + W_{oc}F_{\text{con}t} + b_o\right) \tag{16}$$

$$c_t = i_t \odot \varphi\left(W_{zx}x_t + W_{zh}h_{t-1} + W_{za}F_{\text{ml}_t} + W_{\text{zc}}F_{\text{con}t} + b_z\right) + f_t \odot c_{t-1} \tag{17}$$

$$\hat{c}_t = \beta_t s_t + (1-\beta_t)c_t \tag{18}$$

$$h_t = o_t \odot \tanh(\hat{c}_t) \tag{19}$$

$$s_t = \text{softmax}(W_s h_t) \tag{20}$$

where the parameters W_{ia}, W_{fa}, W_{oa}, and W_{za} represent multilevel feature vectors in the decoder, and the parameters W_{ic}, W_{fc}, W_{oc}, and W_{zc} represent the contextual feature vectors. Here, σ and φ denote the sigmoid activation function and maxout nonlinear function, respectively. β_t is the visual gate at time t, whose range is [0, 1]. When β_t is 1, it means that the generation of the next word only depends on the textual information that has been generated before, while when

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β_t is 0, it means that it only depends on the visual information. This reflects the characteristics of adaptive selection of visual information and contextual information.

As stated in [60], in order to calculate the new visual sentinel gate β_t, an extra element is added to z_t to identify the attention degree between the network and the text characteristics. The value of β_t is calculated as follows:

$$z_t = \omega_h^T \tanh(W_{vt} V + W_{ht} h_t) l^T \tag{21}$$

$$\beta_t = \left[\text{softmax}\left(\text{concat}\left(z_t, \omega_h^T \tanh(W_{st} s_t + W_{ht} h_t)\right)\right)\right]_{k+1}. \tag{22}$$

Here, $V = \{v_1, v_2, \ldots, v_L\}$ represents the original visual feature vector sets, W_{vt}, W_{ht}, and ω_h are learnable weights, and $I \in R^k$ is a vector with all elements set to 1. W_{st} and W_{ht} are weight parameters and W_{ht} is the same as in (21). β_t denotes the attention weight corresponding to the feature vector of the image region, which is the $k + 1$ element of attention distribution over the image features and visual sentinel vector as well.

Finally, cross-entropy loss is utilized to train LSTM. Cross-entropy loss is mainly used to reveal the difference between probability distributions. For captioning tasks, the cross-entropy loss is generally exploited for optimizing the procedure of word prediction to generate a sequence of words. Specifically, all candidate words would be assigned a probability through the softmax function. The goal of cross-entropy loss is to optimize the probability of the predicted word

$$L(\theta) = -\sum_{t=1}^{T} \log\left(p_\theta\left(s_t \middle| s_1^*, \ldots, s_L^*\right)\right) \tag{23}$$

where p_θ represents the predicted word probability at time t by the model, corresponding to the lth word in the label sentence of length L. At each time step, only one word would be generated. θ refers to the softmax classifier parameter.

In the testing process, feedforward information is used to generate a description sentence for a given image. The sentence is initialized with ⟨0⟩ padding and the start flag ⟨start⟩, which is used as input for the prediction of the next word. The selection of predicted words relies upon the beam-search algorithm. The predicted words are then added to the sentence in order, and the process is repeated until the end tag ⟨end⟩ is predicted. Another ending condition is the length of the sentence reaches a predefined maximum.

V. Experiments and Analysis

In this section, we first give a compendious introduction of the classical metrics and the configuration settings used in our experiments. Next, we evaluate MLCA-Net based on different CNNs and analyze the contributions of two attention modules, the multilevel attention module and the contextual attention module, for remote sensing image captioning. Then, we compare MLCA-Net with several representative baselines for RS images and natural images to verify the effectiveness of MLCA-Net. Finally, visualization results, including visual attention maps and example sentences, are given. All experiments are conducted on four datasets.

TABLE II
Data Splitting Strategy for Training–Validation–Testing Sets

Dataset	Training set	Validation set	Test set
UCM-Captions[27]	1,680	210	210
Sydney-Captions[27]	497	58	58
RSICD[28]	8,734	1,093	1,094
NWPU-Captions	25,200	3,150	3,150

A. Metrics

Five metrics, BiLingual Evaluation Understudy (BLEU) [54], ROUGE_L [55], metric for evaluation of translation with explicit ordering (METEOR) [56], consensus-based image description evaluation (CIDEr) [57], and semantic propositional image caption evaluation (SPICE) [58], are used in this article. Given a set of references or GT sentences (i.e., labeled sentences) $S_i = \{s_{i1}, \ldots, s_{im}\} \epsilon S$ and a candidate sentence (i.e. predicted sentence) $c_i \epsilon C$ for an image I_i, the definitions of these five metrics are given as follows.

BLEU [54] measures the similarity between c and S by calculating the co-occurrence of n-grams in them

$$\text{BLEU}_N(c, S) = b(c, S) \exp \sum_{n=1}^{N} \frac{1}{N} \log CP_n(c, S)) \tag{24}$$

where $CP_n(c, S)$ represents the coincidence degree of the global n-gram between c and S, $b(c, S)$ is a penalty term [brevity penalty (BP)] since $CP_n(c, S)$ scores higher in shorter sentences.

Recall-oriented understudy for gisting evaluation (ROUGE) [55] is a summarization evaluation package and includes four measures, among which ROUGE_L is used for the evaluation of image captioning. ROUGE_L estimates the similarity between c and S by calculating F-measure based on the longest common subsequence (LCS)

$$\text{ROUGE}_l(C, S) = \frac{\left(1 + \beta^2\right) R_{\text{lcs}} P_{\text{lcs}}}{R_{\text{lcs}} + \beta^2 P_{\text{lcs}}} \tag{25}$$

where $\beta = P_{\text{lcs}}/R_{\text{lcs}}$ and ROUGE-L is 1 when $c = S$, while ROUGE-L is zero when $\text{LCS}(c, S) = 0$. It has been proven that unigram LCS-based F-measure is as good as BLEU since the composite factors of LCS-based recall and precision are considered.

METEOR [56] measures the similarity between c and S by calculating F_{mean} of precision P_m and the recall R_m through n-gram with a penalty factor Pen

$$\text{METEOR} = (1 - \text{Pen}) F_{\text{mean}} \tag{26}$$

where $F_{\text{mean}} = (P_m R_m)/(\alpha P_m + (1 - \alpha) R_m)$ with α a default evaluation parameter and m the number of matching words in the candidate sentence and the reference sentence.

CIDEr [57] applies term frequency-inverse document frequency (TF-IDF) weights for each n-gram and measures the similarity between c and S by calculating the cosine similarity

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TABLE III
RESULTS OF CNN + LSTM BASED ON DIFFERENT CNNS (UCM-CAPTIONS)

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE	Training time
AlexNet	0.757	0.706	0.655	0.612	0.389	0.695	2.82	0.418	15min
VGG19	0.772	0.721	0.669	0.625	0.401	0.715	2.92	0.438	80min
VGG16	0.768	0.716	0.667	0.621	0.397	0.708	2.91	0.427	60min
GoogLeNet	**0.817**	**0.742**	**0.684**	**0.636**	**0.418**	**0.746**	**3.10**	0.436	90min
ResNet152	0.801	0.736	0.683	0.633	0.411	0.746	3.02	**0.437**	120min

TABLE IV
RESULTS OF CNN + LSTM BASED ON DIFFERENT CNNS (SYDNEY-CAPTIONS)

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE	Training time
AlexNet	0.771	0.678	0.594	0.511	0.632	0.681	2.01	0.377	20min
VGG19	0.787	0.693	0.609	0.525	0.378	0.699	2.04	0.393	90min
VGG16	0.783	0.690	0.603	0.522	0.374	0.696	2.02	0.389	80min
GoogLeNet	0.778	0.671	0.593	0.526	0.381	0.694	2.08	**0.426**	110min
ResNet152	**0.792**	**0.702**	**0.615**	**0.533**	**0.398**	**0.721**	**2.19**	0.374	170min

across n-grams

$$\mathrm{CIDEr}_n(c, S) = \frac{1}{m}\sum_j \frac{g^n(c) \cdot g^n(S)}{\|g^n(c)\|\|g^n(S)\|} \tag{27}$$

where $g^n(c)$ represents a vector of TF-IDF weights and $\|g^n(c)\|$ is the magnitude of $g^n(c)$. Similar to $g^n(S)$ and $\|g^n(S)\|$, CIDEr metric accounts for precision and recall.

SPICE [58] measures the similarity between c and S by calculating similarities between their scene graphs that encode the semantic propositional content of the captions

$$\mathrm{SPICE}(c, S) = F_1(c, S) = \frac{2P(c, S)R(c, S)}{P(c, S) + R(c, S)} \tag{28}$$

where precision $P(c, S) = (|T(G(c))\otimes T(G(S))|/|T(G(c))|)$, $R(c, S) = (|T(G(c)) \otimes T(G(S))|/|T(G(S))|)$, G is the mapping function from the sentence to the scene graph, T is the mapping function from scene graph to n-tuples, and $\otimes$ is a binary matching operator.

With the exception of SPICE, the other four metrics are all based on n-gram matching (generally, n takes a value of 1–4). While CIDEr and SPICE were developed for the evaluation of image captioning, BLEU, ROUGE_L, and METEOR were originally presented for the evaluation of text summaries or machine translations. BLEU is a modified precision metric, and the other four metrics are composite of precision and recall. It has been proven [59] that CIDEr and SPICE have a higher correlation with human judges compared with the other three metrics.

The range of BLEU, METEOR, ROUGE, and SPICE falls between 0 and 1, and the range of CIDEr is between 0 and 5. If the score of the metrics is high, it means that the predicted and the reference sentences marked by humans are highly similar.

B. Experimental Settings

We evaluate MLCA-Net on four datasets, and the data splitting strategy for training–validation–testing sets is 8:1:1, as shown in Table II.

We perform experiments on a workstation with a 3.60-GHz Processor (12 Core), 32-GB RAM, and NVIDIA TITAN Xp GPU. The operating system and deep learning platforms used are Ubuntu 16.04 and PyTorch1.4.0.

The configuration settings are as follows. MLCA-Net resizes the input images to 224 × 224 pixels and sets the hidden nodes' number of LSTM and the word embedding dimension to 512. It sets the initial learning rates of the encoder and the decoder to $1e^{-4}$ and $5e^{-4}$, respectively, which are gradually reduced to 0.8 times the original after every four epochs of iteration. The batch size is set as 32, and the maximum epoch is set to 50. Adam optimizer is used to implement optimization for training the network. Besides, the dropout layer with a probability of 0.5 is added in the image feature extraction stage. In the testing process, the beam-search algorithm is adopted for sentence generation: assume that we set a hyperparameter, beam size, to k. Then, at the first decoding step, we select top k words as a candidate. Each of the words will be further acted as the first k token of candidate output sequences. At each subsequent time step, top k words are selected based on the top k candidate output sequences generated at the previous time stop. Thus beam-search algorithm reaches a balance between accuracy and computational cost through the setting of beam size. In our experiments, the beam size is set to 3. Finally, the length of the generated sentences is set to no longer than 20 words.

C. Results and Analysis

1) Results With Different CNNs: In this section, the performance of basic CNN + LSTM architecture on different CNNs, including AlexNet, VGG16/19, GoogLeNet, and ResNet152, is evaluated. Tables III–VI show results on four remote sensing image captioning datasets. It is shown that a deeper network generally enjoys better performance but needs more time to train. Among the five CNNs, GoogLeNet and ResNet perform better than the other three CNNs on most metrics. The bar chart in Fig. 6 shows that the influence of CNN is really small, except for the UCM-Captions dataset.

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TABLE V
RESULTS OF CNN + LSTM BASED ON DIFFERENT CNNS (RSICD)

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE	Training time
AlexNet	0.684	0.552	0.461	0.393	0.326	0.602	2.07	0.412	80min
VGG19	0.711	0.579	0.481	0.414	0.344	0.616	2.21	0.428	180min
VGG16	0.691	0.567	0.476	0.405	**0.339**	0.613	2.18	0.425	160min
GoogLeNet	0.688	0.545	0.447	0.372	0.332	**0.628**	2.02	0.416	260min
ResNet152	**0.701**	**0.576**	**0.484**	**0.411**	0.335	0.618	**2.25**	**0.428**	340min

TABLE VI
RESULTS OF CNN + LSTM BASED ON DIFFERENT CNNS (NWPU-CAPTIONS)

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE	Train time
AlexNet	0.717	0.590	0.504	0.440	0.320	0.578	1.06	0.265	130min
VGG19	0.729	0.604	0.520	0.456	0.331	0.587	1.07	0.278	250min
VGG16	0.724	0.598	0.514	0.451	0.324	0.578	1.08	0.269	220min
GoogLeNet	**0.731**	**0.610**	**0.526**	**0.463**	**0.337**	**0.596**	**1.10**	**0.278**	320min
ResNet152	0.729	0.604	0.519	0.455	0.331	0.587	1.07	0.277	410min

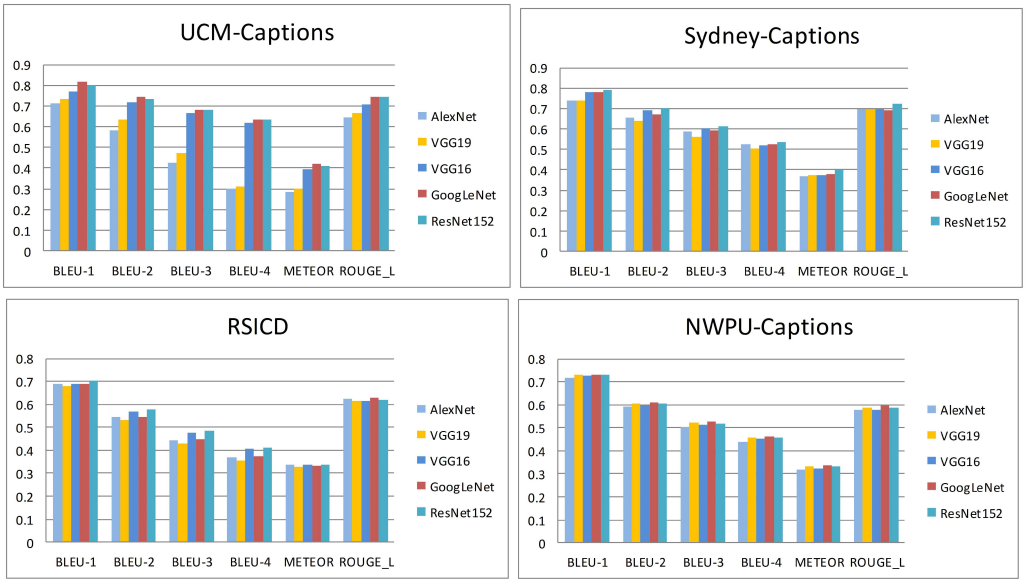


Fig. 6. Comparison results of CNN + LSTM based on different CNNs.

TABLE VII
RESULTS OF ABLATION EXPERIMENTS ON UCM-CAPTIONS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
CNN+LSTM[60]	0.768	0.716	0.667	0.621	0.397	0.708	2.910	0.427
CNN+ML+LSTM	0.797	0.747	0.701	0.658	0.411	0.729	3.086	0.454
CNN+CA+LSTM	0.813	0.753	0.699	0.649	0.435	0.766	**3.381**	**0.486**
MLCA-Net	**0.826**	**0.770**	**0.717**	**0.668**	**0.435**	**0.772**	3.240	0.473

2) Results of Ablation Experiments: The effectiveness of the two attention modules of MLCA-Net is evaluated through ablation experiments. In our experiments, VGG16 is selected as the backbone network after comprehensive consideration of the five metrics and training time.

In our experiments, we select basic encoder–decoder architecture with a visual sentinel [60] as our baseline model and mark it as CNN + LSTM. We mark the baseline modal with ML attention module as CNN + ML + LSTM and the baseline modal with CA module as CNN + CA + LSTM. We mark the baseline modal with both attention modules as CNN + ML + CA + LSTM or MLCA-Net. The results of ablation experiments on four captioning datasets are shown in Tables VII–X, which verify the effectiveness of the ML attention module and CA module of MLCA-Net on four datasets. The improvement of CNN + CA + LSTM is close

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TABLE VIII
RESULTS OF ABLATION EXPERIMENTS ON SYDNEY-CAPTIONS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
CNN+LSTM[60]	0.783	0.690	0.603	0.522	0.374	0.696	2.022	0.389
CNN+ML+LSTM	0.814	0.720	0.631	0.547	0.393	0.709	2.191	0.385
CNN+CA+LSTM	0.813	0.726	0.645	0.575	0.387	**0.726**	2.271	0.400
MLCA-Net	**0.831**	**0.742**	**0.659**	**0.580**	**0.390**	0.711	**2.324**	**0.409**

TABLE IX
RESULTS OF ABLATION EXPERIMENTS ON RSICD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
CNN+LSTM[60]	0.691	0.567	0.476	0.405	0.339	0.613	2.181	0.425
CNN+ML+LSTM	0.726	0.606	0.513	0.437	0.356	0.648	2.421	0.455
CNN+CA+LSTM	0.728	0.603	0.506	0.426	0.333	0.618	2.212	0.427
MLCA-Net	**0.750**	**0.631**	**0.538**	**0.459**	**0.342**	**0.638**	**2.418**	**0.444**

TABLE X
RESULTS OF ABLATION EXPERIMENTS ON NWPU-CAPTIONS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
CNN+LSTM[60]	0.724	0.598	0.514	0.451	0.324	0.578	1.086	0.269
CNN+ML+LSTM	0.736	0.615	0.530	0.466	**0.338**	0.600	1.121	0.282
CNN+CA+LSTM	0.735	0.616	0.534	0.471	**0.338**	0.599	1.119	0.282
MLCA-Net	**0.745**	**0.624**	**0.541**	**0.478**	0.337	**0.601**	**1.164**	**0.285**

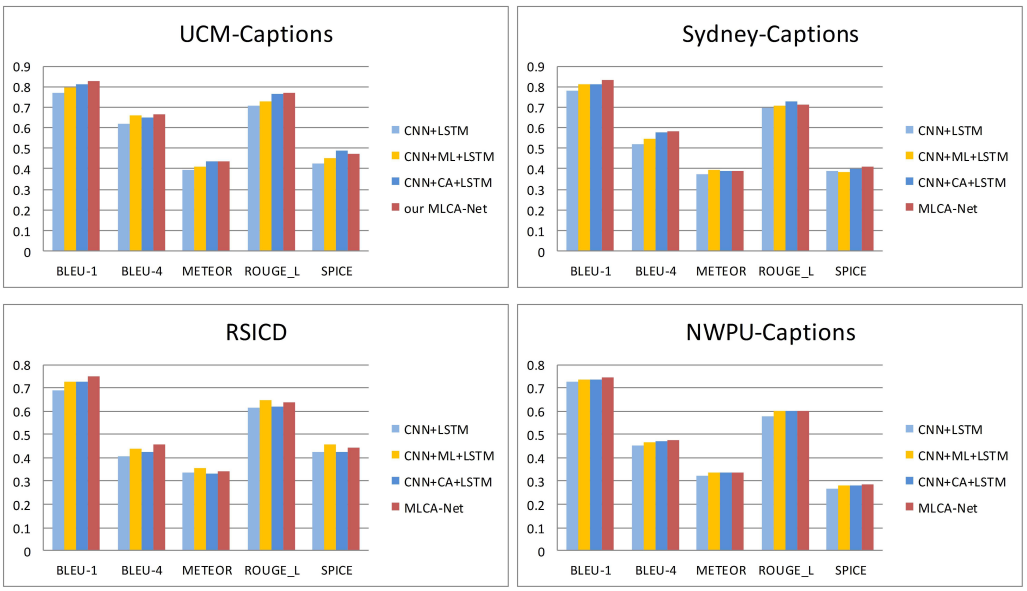


Fig. 7. Comparison results of ablation experiments on four different datasets.

to that of CNN + ML + LSTM, but is slightly different on different datasets. The combination of the two attention modules performs better than that of a separate attention module. The improvements of both attention modules on four datasets are also shown in the bar chart of Fig. 7.

3) Comparison With Baselines for RS Image Captioning: Several available models, CSMLF [45], multimodal method [27], attention-based methods (soft/hard) [28], and FC-Att + LSTM and SM-Att + LSTM [29], are chosen for comparison. Among them, CSMLF employs semantic embedding for cross-modal discrepancy measurement and applies collective representation to improve captioning performance. The multimodal method adopts information from different models, visual and textual, to generate language descriptions. Regarding two similar attribution attention-based models, FC-Att + LSTM and SM-Att + LSTM, the output of different layers acts as high-level attributes to enhance captioning performance.

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TABLE XI
COMPARATIVE RESULTS OF MLCA-NET AND BASELINES FOR RS IMAGES ON UCM-CAPTIONS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
CSMLF[45]	0.437	0.274	0.187	0.122	0.133	0.394	0.223	0.076
Multimodal[27]	0.710	0.598	0.553	0.460	0.343	0.662	2.926	-
Attention(soft)[28]	0.746	0.655	0.587	0.525	0.390	0.724	3.187	-
Attention(hard) [28]	0.817	0.731	0.671	0.618	0.427	0.771	2.996	-
FC-Att+LSTM[29]	0.814	0.751	0.685	0.636	0.417	0.751	2.997	-
SM-Att+LSTM[29]	0.815	0.759	0.694	0.647	0.425	0.763	3.186	-
Our MLCA-Net	**0.826**	**0.770**	**0.717**	**0.668**	**0.435**	**0.772**	**3.240**	**0.473**

TABLE XII
COMPARATIVE RESULTS OF MLCA-NET AND BASELINES FOR RS IMAGES ON SYDNEY-CAPTIONS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
CSMLF[45]	0.602	0.460	0.389	0.343	0.248	0.504	0.757	0.263
Multimodal[27]	0.698	0.613	0.544	0.505	0.361	0.637	2.202	-
Attention(soft)[28]	0.733	0.668	0.623	0.583	0.395	0.714	2.499	-
Attention(hard) [28]	0.762	0.663	0.590	0.527	0.393	0.721	2.183	-
FC-Att+LSTM[29]	0.808	0.717	0.630	0.554	0.411	0.712	2.203	-
SM-Att+LSTM[29]	0.814	0.736	**0.659**	**0.582**	**0.411**	**0.720**	2.302	-
Our MLCA-Net	**0.831**	**0.742**	**0.659**	0.580	0.390	0.711	**2.324**	**0.409**

TABLE XIII
COMPARATIVE RESULTS OF MLCA-NET AND BASELINES FOR RS IMAGES ON RSICD

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
CSMLF[45]	0.577	0.387	0.284	0.223	0.214	0.447	0.531	0.199
Multimodal[27]	0.639	0.477	0.401	0.302	0.292	0.534	2.255	-
Attention(soft)[28]	0.676	0.532	0.434	0.365	0.327	0.612	1.964	-
Attention(hard) [28]	0.668	0.518	0.417	0.341	0.321	0.608	1.794	-
FC-Att+LSTM[29]	0.746	0.625	0.534	0.457	0.340	0.633	2.366	-
SM-Att+LSTM[29]	0.750	0.631	0.538	0.459	0.342	0.638	**2.418**	-
Our MLCA-Net	**0.757**	**0.634**	**0.539**	**0.461**	**0.351**	**0.646**	2.356	**0.444**

TABLE XIV
COMPARATIVE RESULTS OF MLCA-NET AND BASELINES FOR RS IMAGES ON NWPU-CAPTIONS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
CSMLF[45]	0.717	0.590	0.504	0.440	0.320	0.578	1.065	0.265
Multimodal[27]	0.725	0.603	0.518	0.455	0.336	0.591	1.179	0.276
Attention(soft)[28]	0.731	0.609	0.525	0.462	0.339	0.599	1.136	0.285
Attention(hard) [28]	0.733	0.610	0.527	0.464	0.340	0.600	1.103	0.284
FC-Att+LSTM[29]	0.736	0.615	0.532	0.469	0.338	0.600	1.231	0.283
SM-Att+LSTM[29]	0.739	0.617	0.532	0.468	0.330	0.593	1.236	0.276
Our MLCA-Net	**0.745**	**0.624**	**0.541**	**0.478**	**0.337**	**0.601**	**1.264**	**0.285**

Tables XI–XIV report the comparative results of MLCA-Net with the aforementioned four baselines on the four datasets. It shows that MLCA-Net is superior over the baselines in most cases, except for a slight decrease in BLEU-4, METEOR, and ROUGE_L on SYDNEY-Captions, compared with SM-Att + LSTM. We speculate that the decrease might be caused by the small quantity and unbalanced categories of Sydney-Captions. The effectiveness of MLCA-Net is also shown through the bar chart in Fig. 8.

4) Comparison With Baselines Used for Natural Image Captioning: We also compare MLCA-Net with several baselines that achieve satisfying performance on natural image captioning, including Show-Attend-Tell [10], SCST-Att2all [16], and Up-Down Attention [11]. The comparative experiments aim to carry out more comprehensive evaluations of our proposed dataset and model. Among them, Show-Attend-Tell first introduced an attention mechanism into image captioning. Specifically, it uses the attention module on the decoding end to select the required features in the input sequence. SCST-Att2all applies the gradient strategy of RL in model training to tackle the differentiability issue of the metric function. In Up-Down Attention, bottom-up attention and top-down

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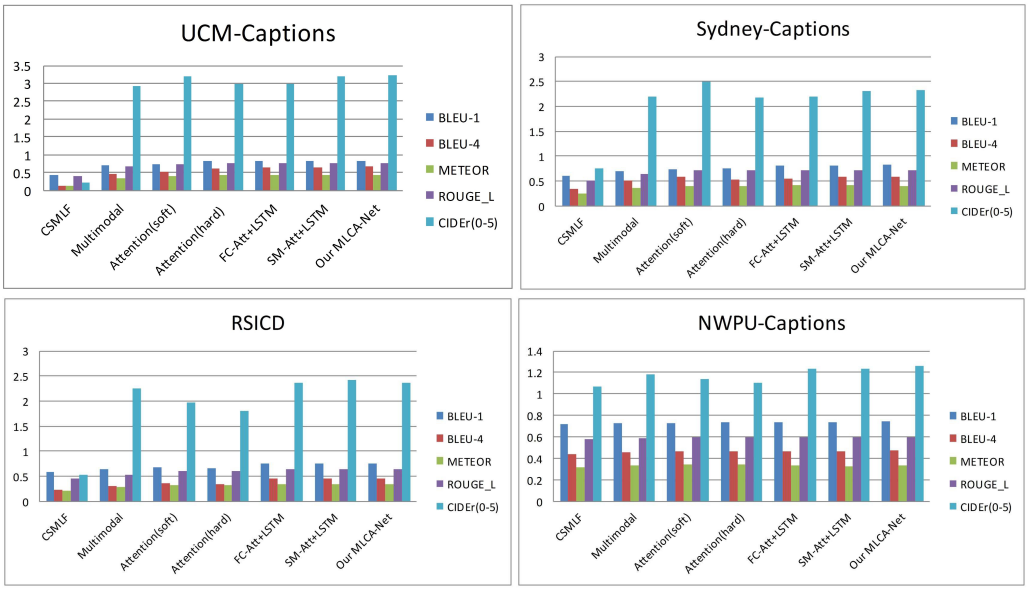


Fig. 8. Comparison results of MLCA-Net with baselines on four datasets.

TABLE XV
COMPARATIVE RESULTS OF MLCA-NET AND BASELINES FOR NATURAL IMAGES ON UCM-CAPTIONS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
Show-Attend-Tell[10]	0.787	0.725	0.670	0.621	0.371	0.702	2.769	-
SCST-Att2all[16]	0.810	0.747	0.694	0.647	0.417	0.750	3.054	-
Up-Down Attention[11]	0.823	0.759	0.707	0.661	0.424	0.763	3.021	-
Our MLCA-Net	**0.826**	**0.770**	**0.717**	**0.668**	**0.435**	**0.772**	**3.240**	**0.473**

attention are employed, respectively, for ROI feature extraction of regions and corresponding weight learning. The two models are combined to boost in-depth understanding of the visual image.

Tables XV and XVI tabulate the comparative results on UCM-Captions and NWPU-Captions. It shows although the three baselines achieve satisfying performance on natural image datasets such as Flickr8K, Flickr30K, and MSCOCO, the generation ability in the remote sensing field needs to be further verified and improved. It also explains the necessity of developing large-scale benchmarks and approaches specifically for remote sensing images.

5) Visualization Results: In this section, some visualization results of MLCA-Net are presented. Fig. 9 shows the visual attention maps of MLCA-Net on four datasets.

For visual attention maps, we use the visual probability to describe the probability of a word that belongs to visual attribute words. In Fig. 9, word and its visual probability for an input image generated by MLCA-Net are annotated at the top and bottom of its visual attention map. Generally, MLCA-Net gives lower visual probability to nonvisual words [e.g., "in" and "the" in Fig. 9(a)] and gives higher visual probability to visual words [e.g., "building" and "playground" in Fig. 9(c)]. However, the same word might be given different visual probabilities in different scenes. For example, the word at the beginning of a sentence is usually given a higher visual probability [e.g., "a" in Fig. 9(b)] even if it belongs to a nonvisual word. The reason is that the model does not have any linguistic context when generating the first word, so visual information is needed to determine singular and plural numbers. Comparatively, if the same word exists in a phrase [e.g., "a" in the phrase "near a playground" in Fig. 9(c)], MLCA-Net gives it a much lower visual probability. Besides, it can be

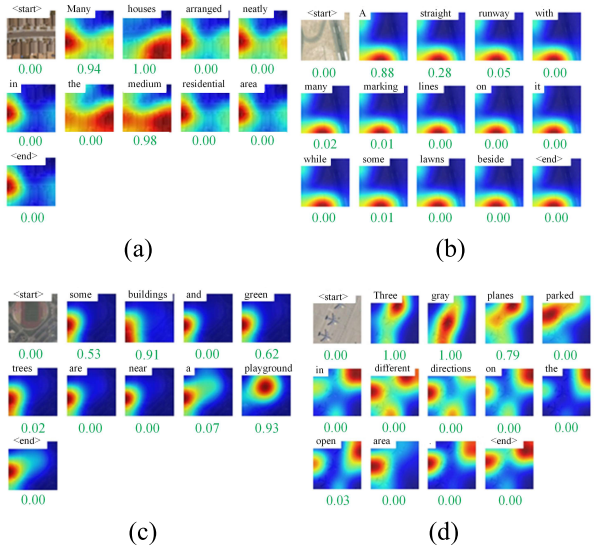


Fig. 9. Visual attention maps of MLCA-Net on four datasets. (a) UCM-Captions. (b) Sydney-Captions. (c) RSICD. (d) NWPU-Captions.

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TABLE XVI

COMPARATIVE RESULTS OF MLCA-NET AND BASELINES FOR NATURAL IMAGES ON NWPU-CAPTIONS

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	ROUGE_L	CIDEr(0-5)	SPICE
Show-Attend-Tell[10]	0.724	0.598	0.514	0.451	0.324	0.578	1.086	0.269
SCST-Att2all[16]	0.735	0.616	0.533	0.471	0.330	0.589	1.149	0.279
Up-Down Attention[11]	0.738	0.619	0.536	0.472	0.338	0.600	1.140	0.281
Our MLCA-Net	**0.745**	**0.624**	**0.541**	**0.478**	**0.337**	**0.601**	**1.164**	**0.285**

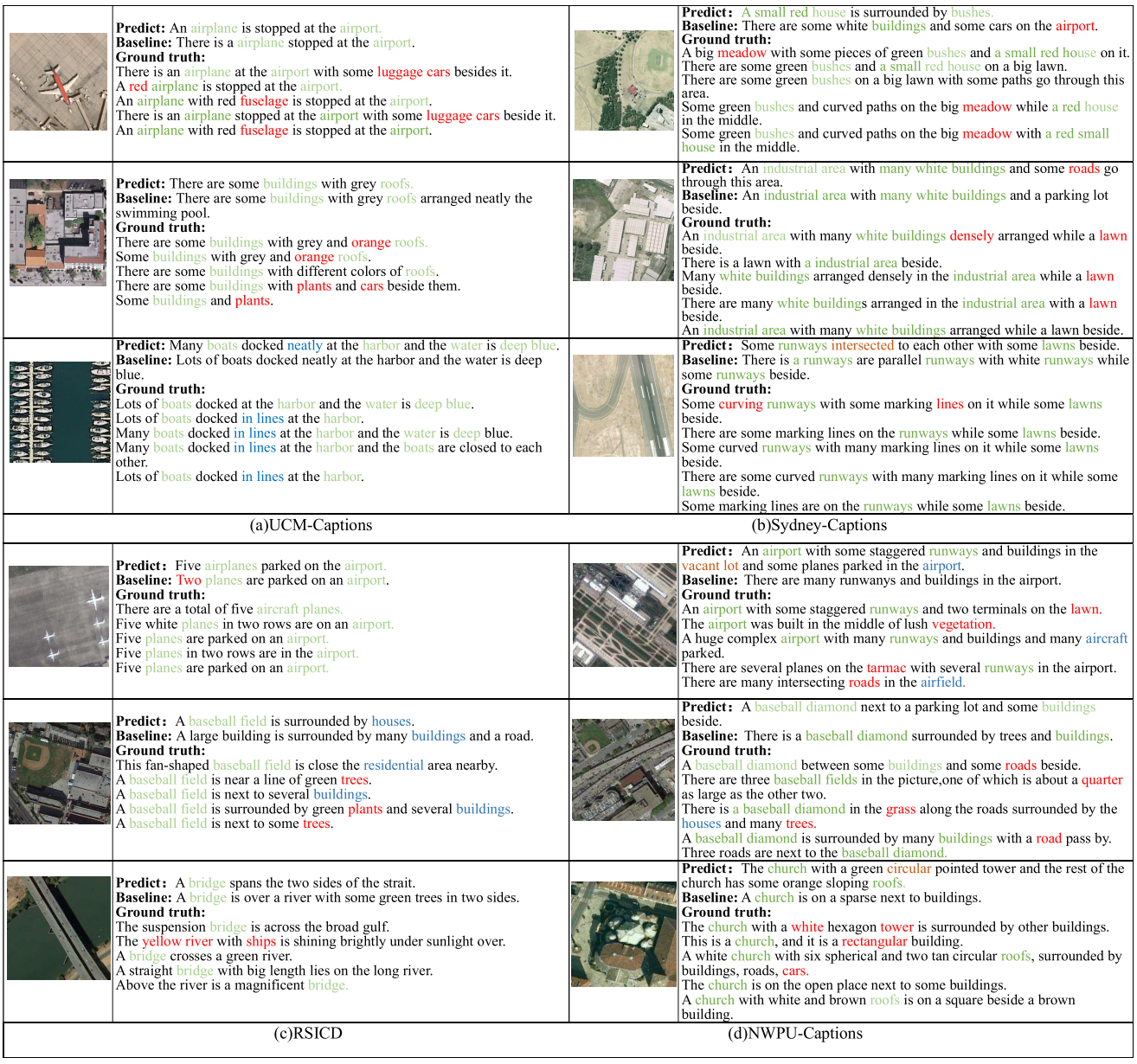


Fig. 10. Example sentences generated by MLCA-Net on four datasets. (a) UCM-Captions. (b) Sydney-Captions. (c) RSICD. (d) NWPU-Captions.

noticed that the visual probability of subsequent words is close to "0" because LSTM will gradually forget the previous visual information during the decoding process. The automatic adjusting of visual probabilities reflects the self-adaptability for visual attention of MLCA-Net.

Fig. 10 shows the example sentences generated by MLCA-Net on four datasets. For each dataset, we give three different scenes. Different colors are used to mark correctness, error, and lexical novelty. Specifically, correct objects and attributes are marked in green, and novel words are marked in orange. For example, the predicted "vacant lot" in the first row and the predicted "circular" in the third row in Fig. 10(d), which are correct yet not present in GT sentences, are marked in orange. Synonyms are marked in blue. For example, the predicted

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"airport" in the first row is synonymous with "aircraft" and "airfield" in GT sentences, so the three words are marked in blue. Besides, predicted wrong words or unpredicted words are marked in red. For example, "roads" in the second row of Fig. 10(b) is a wrong word and is thus marked in red. The words "yellow river" and "ships" in GT sentences of the third row in Fig. 10(c) are not predicted by the proposed MLCA-Net, hence marked in red.

It shows that the predicted sentence of MLCA-Net generally keeps semantic consistency with GT. For example, "grey roofs" in the second row of Fig. 10(a) and "white buildings" in the second row of Fig. 10(b) accurately describe the objects in an image. The word "five" in the first row of Fig. 10(c) and "a small red" in the first row of Fig. 10(b) exactly describe the attribute information of an image. The phrase "spans the two sides of" in the third row of Fig. 10(c) precisely describes the spatial relationship in an image. Besides, MLCA-Net tries to express remote sensing scenes with synonymous or novel vocabularies, for example, "vacant lot" in the first row of Fig. 10(d), which does not appear in the reference sentences.

However, it should be admitted that there also exist some inaccurate or wrong descriptions, as marked in red (e.g., "roads" in the second row of Fig. 10(b). This may be due to the fact that certain words occur simultaneously in the training set. For instance, buildings and roads frequently emerge in one image, so the words "building" and "road" are always annotated in one reference sentence, which results in the incorrect classification of MLCA-Net.

We also give the results presented by a baseline method [60]. We noticed that although, in most cases, the generated captions are correct, wrong descriptions exist. For example, for the sentence "Two planes are parked on an airport" in the first row of Fig. 10(c), the number should be five. For the sentence "A large building is surrounded by many buildings and a road" in the second row of Fig. 10(c), "baseball field" is not predicted. For the sentence "There is a runways are parallel runways with white runways while some runways beside." in the third row of Fig. 10(b), there exists obvious grammar mistake.

In a word, it shows that MLCA-Net should be improved in sentence coding ability in high-frequency words and the ability to discriminate the phenomenon of "same spectrum different thing" and "same thing different spectrum" in remote sensing image scenes.

VI. Conclusion

One of the contributions of this article is to present the new benchmark dataset NWPU-Captions. NWPU-Captions is superior over existing datasets in terms of data volume, category variety, and description richness, which can be served to develop novel approaches. Another contribution is to propose a novel encoder–decoder architecture, MLCA-Net, by exploiting multilevel attention and contextual attention. We aim to provide feasible solutions to tackle the challenges of scale discrepancy and category ambiguity faced by remote sensing image captioning. The performance of MLCA-Net is evaluated on existing datasets and NWPU-Captions with several available baselines, either for captioning on remote sensing images or on natural images. Our future work includes sequential data description and cross-modal retrieval of multisource data.

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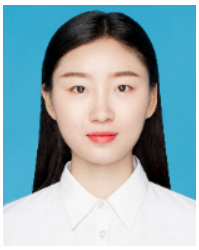
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